

Weekly CT-02

Cambridge International AS and A Level Computer Science 9618

Networks including the internet

Time: 30 minutes

Total Marks: 30

Answer all the questions.

1. A company has a LAN (local area network).

[4]

(a) Give two benefits of connecting computers to a LAN.

[2]

(b) Give two characteristics of a LAN.

[2]

2. The following incomplete table contains four network devices and their descriptions.

[7]

(a) Complete the table by writing the missing devices and missing descriptions.

[4]

Device	Description
	Receives and sends data between two networks operating on the same protocol
Wireless Network Interface Card (WNIC)	
	Restores the digital signal so it can be transmitted over greater distances
Wireless Access Point (WAP)	

(b) Describe three differences between fibre-optic cables and copper cables.

[3]

3. A company uses cloud computing.

[6]

(a) Define cloud computing.

[1]

(b) State what is meant by a public cloud and a private cloud.

[2]

Public cloud: _____

Private cloud: _____

(c) Give two benefits and one drawback of using cloud computing.

[3]

Benefit 1: _____

Benefit 2: _____

Drawback: _____

4. A Local Area Network (LAN) uses a bus topology.

[4]

Describe how Carrier Sense Multiple Access/Collision Detection (CSMA/CD) is used in a bus network.

5. A networked CCTV system transfers data over a network. Each device on the network has an IP address.

[5]

Complete the description of IP addresses.

An IPv4 address contains _____ groups of digits. Each group is represented in _____ bits and the groups are separated by full stops.

An IPv6 address contains _____ groups of digits. Each group is represented in _____ bits. Multiple groups that only contain zeros can be replaced with a _____ .

6. A school has a Local Area Network (LAN). The LAN connects to the internet using a router. [3]

Describe the function of a router in a network.

7. How long did you study for this short exam?

[0.5]

Hour = _____

8. Thank you for participating in the exam honestly.

[0.5]